RARL AI PRACTITIONER SERIES

VOLUME 1

Write Like a Researcher

A Field Guide to AI-Assisted Scientific Writing



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RARL
Robotics & Automation Research Lab.

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Companion volume: *Train Like an Engineer: A Guide to Reliable AI Experiments.*

This guide is written for people who use AI systems to draft, edit and review scientific work. It does not replace a venue's author instructions, an institution's ethics policy or a supervisor's judgement. Where they differ, they win.

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CHAPTER 0

Before You Begin

An AI assistant will hand you a fluent paragraph in seconds. It will not tell you whether the paragraph is true or even needed. This guide is about the checking that stays with you.

Who this guide is for

You use an AI assistant to help with research writing: a student drafting a first paper, an engineer turning experiments into a journal article, a clinician or analyst reporting a study, a supervisor editing someone else's work. You have probably noticed the pattern. The sentences are smooth, but the draft reads like a project report, repeats its numbers, overclaims, wanders, and sometimes cites papers that do not exist.

Below are the rules I use to fix that, and the wording to make an AI follow them. Each rule comes with its reason, so you can tell when your venue or field needs an exception.

Chapters 1 to 8 walk through the manuscript section by section. Chapter 9 turns the rules into briefs and review passes for an AI assistant. Chapter 10 is a checklist and a master prompt you can paste in.

One method, many fields

These rules started out on a robotics paper and a machine-vision paper, but nothing in them depends on drones or images. Substitute your own "system under study" and "evaluation scenarios":

Field	System under study	Evaluation scenarios
Robotics & control	Vehicle, manipulator, controller	Environments, disturbances, trajectories
Machine learning	Model or training method	Datasets, domain shifts, ablations
Health & clinical	Intervention, diagnostic tool	Cohorts, sites, follow-up windows
Natural sciences	Material, organism, process	Conditions, concentrations, time points
Social & policy	Programme, instrument, population	Settings, subgroups, comparison periods

THE IDEA BEHIND EVERYTHING ELSE

A paper does not answer "how was the software built?" unless the software is the contribution. It answers: **what problem was studied, why was this approach right, how was it tested, what evidence came out, and what does that evidence show?**

CONVENTIONS ARE DEFAULTS

Some rules here (third person, no citations outside the introduction and literature review, one typeface in figures) are strong defaults, not laws. Your target journal's instructions always take precedence.

CHAPTER 1

A Paper Is a Story

A paper is one argument told in order. This chapter sets out the chain I hold every manuscript against.

Each section exists to hand the reader to the next one. If a reader asks *"but why did they do that?"* and the paper has not answered, the paper has a hole.

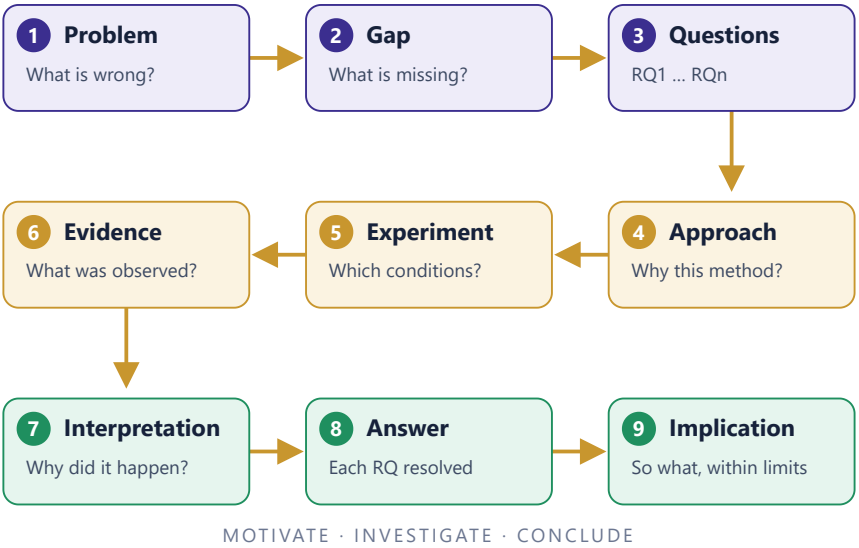


FIG 1 *The reasoning chain of a scientific paper. Violet nodes motivate the work, gold nodes carry it out, green nodes draw conclusions. Every section should strengthen exactly one link.*

Ask for five roles, one at a time

Asking an AI to "improve" a draft gets you polish. It catches more if you make it play one role per pass: the **researcher** (is the question real, is the logic sound?), the **writer** (is each sentence precise?), the **reviewer** (where would a skeptic push back?), the **domain expert** (are terms and magnitudes plausible?) and the **editor** (what is repeated, padded or off topic?).

If I had to compress the whole guide into one sentence: put scientific contribution ahead of implementation detail, interpretation ahead of description, and evidence ahead of claims.

The WH test

Before a draft is finished, a reader should be able to find the answer to each of these without hunting. If a question arises naturally while reading and the text does not answer it, the manuscript is incomplete.

Ask	The manuscript must say
What / Why	What was studied, and why it mattered
Where / When	Setting, site, period, where relevant
Who / which	The subjects, system, dataset or model; which methods and parameters were chosen
How / how much	How it was investigated; how much data, material or compute was used
What happened	What was observed, and what it meant
Why that result	Why it occurred, and what follows from it

CHAPTER 2

Voice, Tone & Tense

Most of what goes wrong in AI-drafted papers is voice: the text sounds like a software report. Settle these decisions first.

Four decisions, made once

Third person

No *I, we, our, my.* Write "The study investigated…", "The results indicated…". The authors stay out of the prose.

Past tense for what was done

Methods, experiments, data, results: "The model was trained…". Completed work is past.

Present tense for what is true

Established facts, definitions and paper structure: "Rotor thrust scales with…", "Section 4 presents…".

Voice follows clarity

Active when the actor matters ("The analysis identified three factors"), passive when the object does ("Samples were incubated at 37 °C"). Never mechanically all one.

Words to retire

Adjectives that sell rather than measure erode trust. Remove *very, excellent, amazing, powerful, highly effective, revolutionary*. Use *significant* only when a statistical test earned it. Claims must be proportional to evidence.

A paper is not a project report

AI assistants drift toward describing *how the code was run*. Papers describe *what was scientifically done and why.*

READS LIKE A REPORT

"The Python function `calculate_velocity()` was called and the script printed the result."

"The code used a CNN function to classify the images."

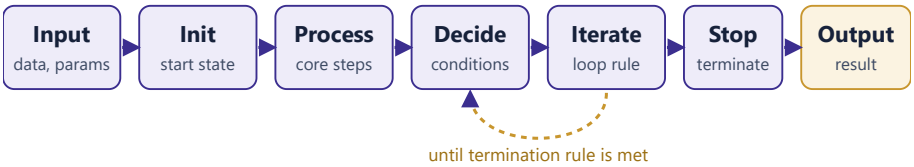
READS LIKE A PAPER

"Velocity was estimated from the measured positional states."

"A convolutional network extracted hierarchical spatial features and performed classification."

Avoid file names, function calls, variable names and “the following code“. If a procedure has several logical steps, present it as an **algorithm** or an equation, not as source.

A PAPER-GRADE ALGORITHM STATES SEVEN THINGS



Test: could a reader reproduce the logic with no access to your source code? If not, the algorithm block is missing a step.

FIG 2 *The anatomy of an algorithm block. It replaces pages of code description with seven stated elements.*

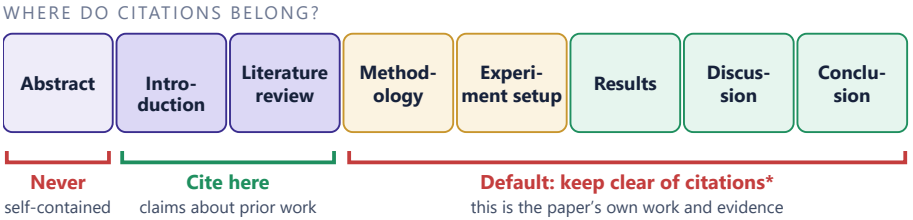
KEEP TERMS CONSISTENT

Pick one term and keep it: *drone / UAV / quadrotor*, *cohort / sample / group*, *model / network / classifier*. Define every abbreviation once at first use, then use it consistently. Keep units uniform (m/s *or* km/h, not both) and never let one symbol mean two things.

CHAPTER 3

Anatomy of a Paper

What each section is for, and where citations are allowed to appear.



*** Venue first.** If a later claim truly needs outside attribution (borrowed metric, published protocol, dataset licence), follow the journal's rules and cite it. Never add a citation to look scholarly.

FIG 3 *The eight standard sections and the default citation zones. Intro and literature review carry the weight of prior work; the abstract stands alone.*

The four-beat rhythm

No section should feel stapled in. Each runs the same four beats, ending in a hand-off. At paragraph scale the rhythm is **Topic → Evidence → Interpretation → Connection.**

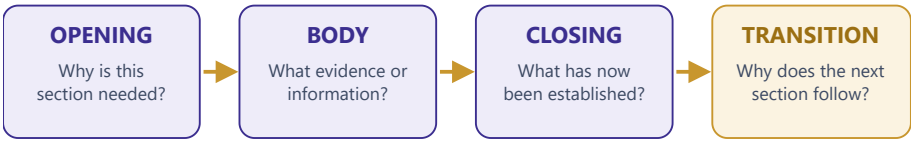


FIG 4 *The four beats of every major section. Method follows from the gap, results from the design, discussion from the results. An abrupt ending is the most common structural flaw in AI drafts.*

CHAPTER 4

Opening Moves

The abstract, introduction, research questions and literature review. This is where a reviewer makes up their mind.

The abstract: six moves, no citations

It must be fully self-contained: a reader who sees only the abstract should know what was investigated and found.

#	Move	One sentence that…
1	Background	names the problem
2	Gap	says what is unresolved
3	Objective	states what the study set out to do
4	Method	summarises the approach, without detail
5	Result	gives the main finding (numbers only if essential)
6	Implication	states what the finding means

No references. No unexplained acronyms. No claims the paper does not support.

The introduction is a funnel

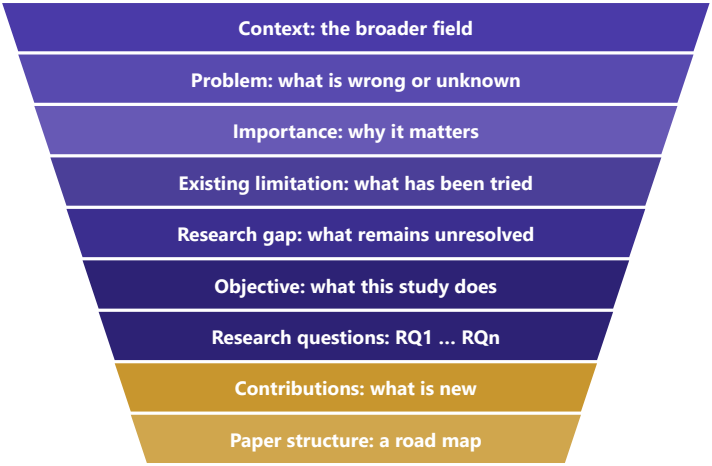


FIG 5 *The introduction narrows from the broad field to the specific contribution. It must not turn into a literature review or a software tour.*

Research questions that earn their place

Label them **RQ1, RQ2, RQ3**. Each must be specific, researchable, answerable by *your* methodology, and later answered in the paper. A question introduced "for appearance" is a trap you set for yourself, a reviewer will look for its answer.

VAGUE

RQ1: Does the proposed method work well?

RESEARCHABLE

RQ1: How does the proposed method change prediction error relative to the baseline under shifted operating conditions?

End the introduction with a road map

"The remainder of the paper is organised as follows. Section 2 reviews related work and identifies the gap. Section 3 presents the methodology…" Describe only sections that exist.

The literature review: synthesise, don't list

A PARADE

"Author A did X. Author B did Y. Author C did Z."

AN ARGUMENT

"Three families of approach exist. The first… but fails when…; the second resolves this at the cost of…; none addresses…"

Group by theme, method, strength, limitation, contradiction and trend. Every paragraph should push the reader one step closer to **the gap**, and the review should end by saying exactly where the present study sits.

CHAPTER 5

Methods Others Can Reproduce

A methods section has to let someone else repeat the work. It does not have to explain your code.

What a reader needs to reproduce your work

- [ ] Research design and its rationale
- [ ] The system under study, with only the specifications that matter
- [ ] Data or experimental input, and its source
- [ ] Variables and parameters
- [ ] Model or method configuration
- [ ] Evaluation metrics, and why they fit
- [ ] Baselines and comparison methods
- [ ] Number of trials and randomisation
- [ ] Environment: hardware, simulator or laboratory conditions
- [ ] Initial conditions and protocol

Choose a primary approach

If you tried several methods, do **not** give them equal space. Present the strongest as the primary methodology, explain why it was chosen, and treat the rest as *baselines, comparators, ablations or benchmarks.* An implementation diary is not a method section.

> **THE COMPARISON QUESTION**
>
> A comparison must answer: **Which approach performed better, under what conditions, and why?** Judge on performance, robustness, cost, generalisation and fit to the problem, not on an inventory of what each tool does.

Describe the system, not the stack



Show the scientific information flow; never class diagrams, file names or code variables.

FIG 6 *A generic system-architecture figure. Replace each block with your own components; keep the flow of information, drop the software internals.*

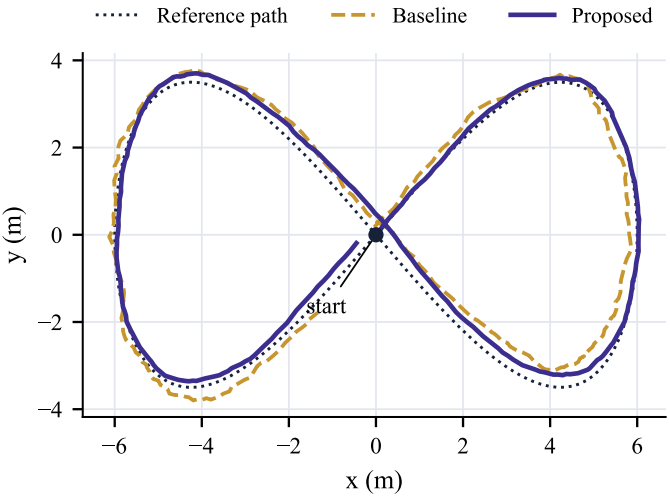


FIG 7 *Reference path and the paths followed by two controllers in one evaluation scenario. A scenario figure should show what the scenario tests. Sample data, synthetic.*

Equations and parameters

- **Introduce equations before using them**, define every variable, and include only those that serve the argument.
- **Report hyperparameters once, compactly** (a table), not repeatedly through the text. Examples: learning rate, batch size, epochs, optimiser, network depth, controller gains, sampling time, thresholds, simulation step, population size.

- **Name architectures explicitly.** "A neural network was used" is not a method. State input size, layers, filters, kernels, strides, activations, pooling, output and parameter count where relevant.
- **Only scientifically relevant specifications.** Do not paste a datasheet or list software merely because it was installed.

> **IF THE STUDY USES HARDWARE OR SIMULATION**
>
> Where the study uses physical or simulated systems, describe: the architecture, sensors, actuators and computing platform; the simulator, time step, duration and control frequency; and **each evaluation scenario**, its purpose, geometry or conditions, start and target states, disturbances, the challenge it poses, and *what scientific question it tests.* A screenshot is not a description. Use the same final system representation in every figure.

CHAPTER 6

Results Without the Noise

Say what happened, once, and keep the interpretation for the discussion. Claims should not run ahead of the evidence.

Results versus discussion

Results ask

What happened? What was observed, how large was the effect, under which condition, how consistent, and what evidence supports it.

Discussion asks

Why, and what does it mean? Mechanism, comparison with prior knowledge, implication, limitation. New results do not appear here.

Say each number once

If a table or figure already shows 95.2% versus 92.1%, the text should not recite them again. Spend the sentence on meaning.

NUMBER RECITAL

"Method A achieved 95.2%, while Method B achieved 92.1%, and Method C achieved 90.4%..."

MEANING FIRST

"Method A consistently outperformed the baselines across all conditions, indicating better generalisation (Table 3)."

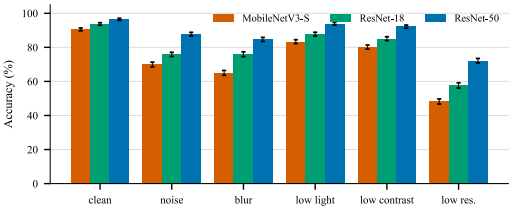


FIG 8 *Accuracy of three models on clean and corrupted photos; error bars are 95% bootstrap intervals over images. Real photos (Imagenette); all numbers are illustrative, not measured.*

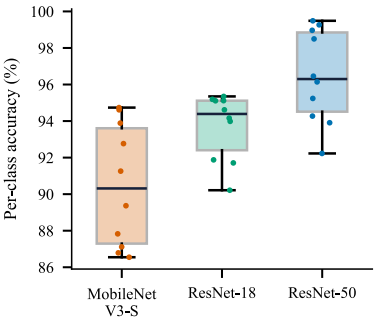


FIG 9 *Per-class accuracy; each dot is one of the ten classes. Real photos (Imagenette); all numbers are illustrative, not measured.*

The bars answer "which model is better, under which condition"; the box plot shows the spread across classes behind each average. Both state what the interval or box means, which is the part readers most often have to guess.

Statistical honesty

Where applicable report **mean and standard deviation, confidence intervals, number of trials, significance tests and variability.** Never claim improvement from a single run. Do not paste raw logs; select what answers the research questions.

Match the verb to the evidence

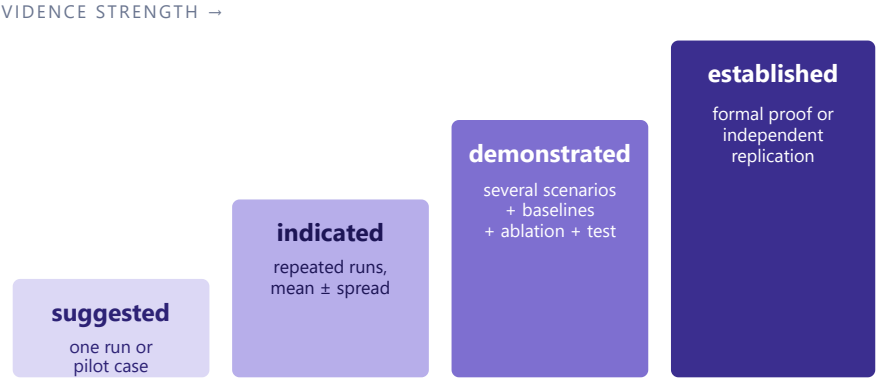


FIG 10 *A claim ladder. One simulation is "suggestive", not "statistically robust"; simulation results do not prove real-world performance.*

WORDING

✓ "The simulation results indicated…"

✗ "The system was proven to work in all real-world conditions."

CHAPTER 7

Figures, Tables & Captions

A figure should answer one question. If you cannot say which question, cut it.

Choose the simplest chart that tells the truth

	The question	Use
	How does a quantity change over time or epochs?	Line plot / time series
	Which method is better?	Bar, box or violin plot (with spread)
	How is a quantity distributed?	Histogram / density plot
	How sensitive is the result to two parameters?	Heat map or line plot
	Do two variables relate?	Scatter plot with fit
	How does the system work?	Architecture or workflow diagram

Avoid decorative 3D, overcrowded plots, tiny labels, redundant graphs and ambiguous legends. Use **one typeface** in every figure (Times New Roman unless the venue says otherwise) and a resolution suitable for print.

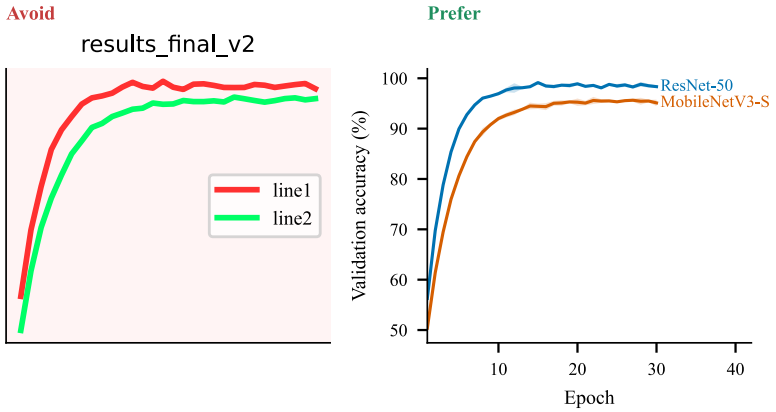


FIG 11 *The same real training curves drawn carelessly and carefully. The careful version labels axes with units, names lines directly, and shows seed spread. Real photos (Imagenette); all numbers are illustrative, not measured.*

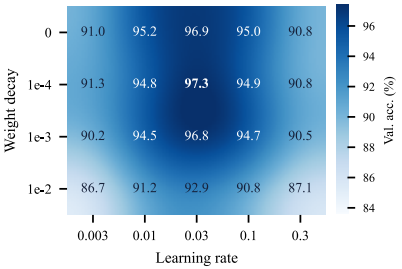


FIG 12 *Linear-probe validation accuracy over a learning-rate × weight-decay grid (ResNet-18 features, 30 epochs). Best cell in bold. Real photos (Imagenette); all numbers are illustrative, not measured.*

Parameter sensitivity is the standard use for a heat map. Keep the number in each cell, pick one sequential colour scale, and say which cell is the setting used in the main experiments.

The anatomy of a good caption

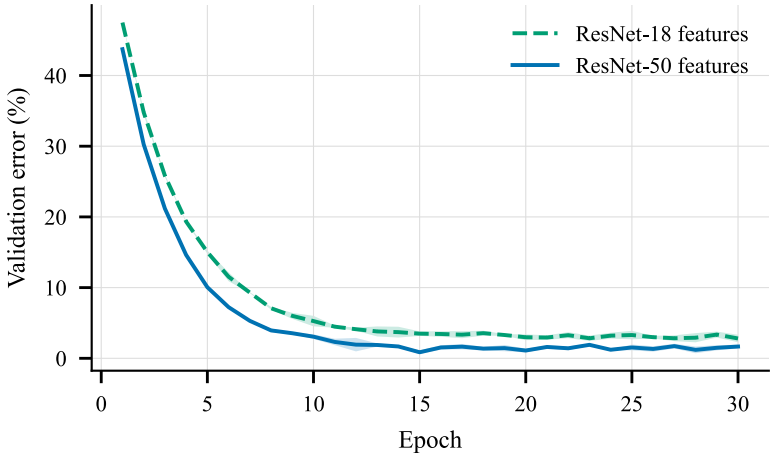


FIG 13 *Validation error against epoch for linear probes on ResNet-18 and ResNet-50 features (mean of three seeds; band is ±1 SD). Check whether the seed bands separate before claiming a difference. Real photos (Imagenette); all numbers are illustrative, not measured.*

That caption does four jobs. It says **what is plotted** (validation error against epoch), the **conditions** (Imagenette, three seeds), what the **shading and lines** mean (±1 SD), and **what to notice** (whether the seed bands separate, which decides whether a difference can be claimed). Compare it with the useless "Simulation result."

In the text, never write "Figure 7 shows the result." Introduce the figure *before* it appears and say **why it matters**: "Figure 7 compares trajectories under both conditions, allowing the stability of the proposed controller to be assessed."

Tables

TABLE 1 *Illustrative top-1 accuracy (%) for three models on Imagenette, clean and under five corruptions, no fine-tuning. Clean accuracy shows a 95% bootstrap interval over images. Best in bold, second underlined. Latency is per image.*

Model	Params (M)	ms / img	clean	noise	blur	low light	low contrast	low res.	mean corr.	ECE (%)
MobileNetV3-S	2.5	0.41	90.5 [89.6, 91.5]	69.9	65.0	83.4	80.3	48.2	69.3	8.1
ResNet-18	11.7	0.62	93.7 [93.0, 94.5]	75.8	75.7	87.7	85.1	57.7	76.4	8.7
ResNet-50	25.6	1.48	**96.5** **[95.9, 97.0]**	**87.7**	**84.7**	**93.9**	**92.2**	**72.2**	**86.2**	7.4

Use a table when structured information is clearer in rows and columns: specifications, hyperparameters, scenarios, metrics, comparisons. Give each a clear title, meaningful headings, consistent units and decimal precision, and notes where needed. A table does not exist because data *can* be tabulated.

The table above follows the usual conventions: one row per system, the clean result first with its interval, best in bold and second underlined, units in the caption, and no vertical rules.

Prompting an AI to draw a scientific figure

When an image-generation system produces an architecture diagram, scenario picture or model render, vague prompts yield decorative nonsense. Specify the figure like an engineer.

Specify	Example
Subject & purpose	"Block diagram of the sensing → estimation → control pipeline, to show information flow"
Viewpoint	Orthographic, top-down; or isometric at 30°
Objects & positions	Which items, where, in what order
Labels	Exact label text and placement; scientifically relevant labels only
Style	Clean vector look, flat lighting, white background, publication quality
Typeface	**"Use Times New Roman for all labels, annotations, legends and textual elements."**
Exclusions	No watermarks, decorative text, fake equations, UI chrome or artistic distortion

CHECK GENERATED FIGURES

The generated image must match what you actually did. Never submit a generated picture of a system, setup or result that differs from the real one, and follow your venue's policy on AI-generated figures.

CHAPTER 8

Figures a Paper Needs

Seven kinds of figure turn up again and again in strong papers. Each one answers a different question for the reader. This chapter shows an example of each, built for this book, and explains when to use it, how to make it and what to avoid.

Think of a paper's figures as a set that covers the whole argument: **what the system is** (method and system diagrams), **how it runs** (the algorithm), **how it performs** (the trade-off plot), **why it behaves that way** (the embedding map), and **what that looks like on real data** (the prediction grid and the annotated screenshot). If one of those questions has no figure, the reader has to build the picture alone from the prose.

1. The trade-off scatter: where every system lands

When two metrics pull against each other, a single number per system hides the story. A scatter with one point per system shows the trade-off directly and lets the reader see which systems dominate.

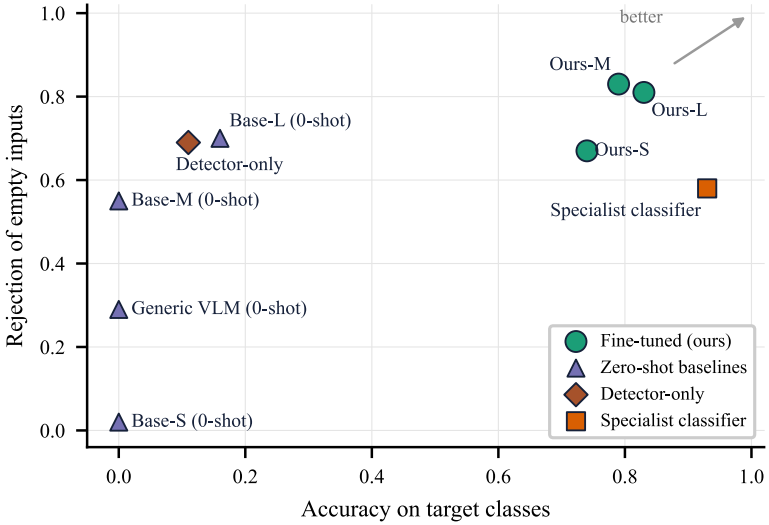


FIG 14 *Accuracy on target classes against rejection of empty inputs. Marker shape and colour encode the kind of system; each point is labelled directly. Fine-tuned models sit in the upper right; zero-shot baselines cluster near zero accuracy. Illustrative data.*

USE IT WHEN

you compare several systems on two competing metrics, such as accuracy and false-alarm rate, or quality and speed.

HOW TO BUILD IT

- One point per system, labelled next to the point rather than in a legend.
- Shape and colour encode the *kind* of system (ours, baseline, specialist); the legend explains only those groups.
- Mark the better direction with a small arrow or a note.
- Keep both axes on the same 0–1 scale when the metrics are rates.

AVOID

- Labels that overlap points; move them by hand.
- More than about four groups.
- Joining points with lines when they are not a sequence.

2. The algorithm block: the procedure without the code

Chapter 2 argued that a paper describes procedures, not source code. The algorithm block is how. Readers should be able to reimplement the step from the block alone.

Algorithm 1 Batch prediction and scoring

Require: inputs $\mathcal{D}$; label set $\mathcal{C} = \mathcal{C} \cup \{\textsc{None}\}$; prompt π; model f_θ
Ensure: predictions $\{(\hat{y}_i, c_i)\}$ and metrics
for all inputs $i \in \mathcal{D}$ **do**
  $v_i \leftarrow$ visual tokens of input i from the frozen encoder
  $s_i \leftarrow f_\theta(v_i, \pi)$ ▷ greedy decoding, 64-token budget
  if s_i parses as `Label|Score` and Label $\in \mathcal{C}$ **then**
    $(\hat{y}_i, c_i) \leftarrow \textsc{Parse}(s_i)$
  else $(\hat{y}_i, c_i) \leftarrow (\textsc{Malformed}, \emptyset)$ ▷ counted as wrong, never dropped
  end if
end for
split $\mathcal{D}$ into target inputs S and empty inputs E by ground truth
return accuracy on S, rejection rate on E, malformed rate

Typeset in LaTeX with `algorithm` + `algpseudocode`: `\Require`, `\Ensure`, `\ForAll`, `\If … \Else … \EndIf`, and `\Comment{…}` for the right-hand notes. Refer to it in the text ("Algorithm 1 summarises…") and point back to the equation the metrics come from.

USE IT WHEN

a step has a loop, a decision or a failure case that prose would blur, e.g. inference, matching, training schedules.

HOW TO BUILD IT

- Start with **Require** (inputs) and **Ensure** (outputs).
- One action per line; math notation for quantities; small caps for named sub-routines.
- Use right-aligned comments for the one detail a reviewer will ask about, such as the decoding budget.
- Say what happens to failures (here: malformed outputs count as wrong).

AVOID

- Pasting Python with variable names.
- Lines that hide three operations.
- Notation that the text never defines.

3. The prediction grid: show the reader real cases

Numbers say how often a system is right. A prediction grid shows *what* right and wrong look like. Rows are conditions (sites, datasets, lighting); columns are case types chosen by a rule.



FIG 15 *Prediction grid. Rows: two capture conditions; the second is the same scene type at night. Columns: case types chosen by rule (all correct, models disagree, all wrong). Border colour gives the overall outcome; each model's prediction is printed below in green (right) or red (wrong). Real photos (Imagenette); numbers are illustrative.*

USE IT WHEN

you want readers to see typical successes, disagreements and failures, especially across conditions.

HOW TO BUILD IT

- Fix the case types first, then pick items by a seeded rule, not by taste.
- Same layout in every row so the eye compares down the columns.
- Put ground truth in the title and each model's prediction under the image, coloured by correctness.
- State the selection rule in the caption.

AVOID

- Showing only successes.
- Cropping out the context that explains a failure.
- Faces, licence plates or private scenes without consent or blurring.

4. The embedding map: why the model behaves as it does

t-SNE or UMAP plots are easy to over-read, so use them to make one point clearly. The strongest version is a **before / after pair** on identical axes and settings.

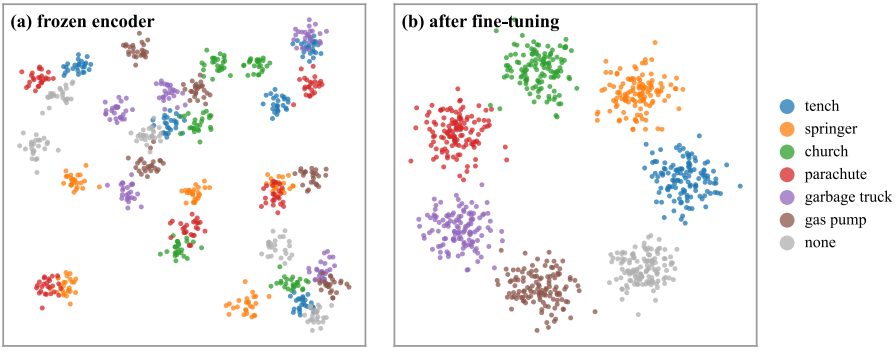


FIG 16 *Feature embeddings of the same inputs (a) from a frozen encoder and (b) after fine-tuning, same perplexity and seed. Classes that were split into many islands become one compact cluster each. Illustrative data.*

USE IT WHEN

you claim that training changed the representation, or that two domains overlap or separate.

HOW TO BUILD IT

- Same inputs, same projection settings, same seed for both panels.
- Small, semi-transparent markers; legend outside the plot.
- Panel labels (a), (b) in the corner instead of titles.
- Say in the caption that distances between clusters are not meaningful in t-SNE.

AVOID

- Reading cluster distances as similarities.
- A plot title on top of a caption that repeats it.
- Default colours that clash with the rest of the paper.

5. The method diagram: the idea in one picture

The method figure is often the one readers remember. Show data flowing from inputs to the loss, with the components that learn and those that do not clearly told apart.

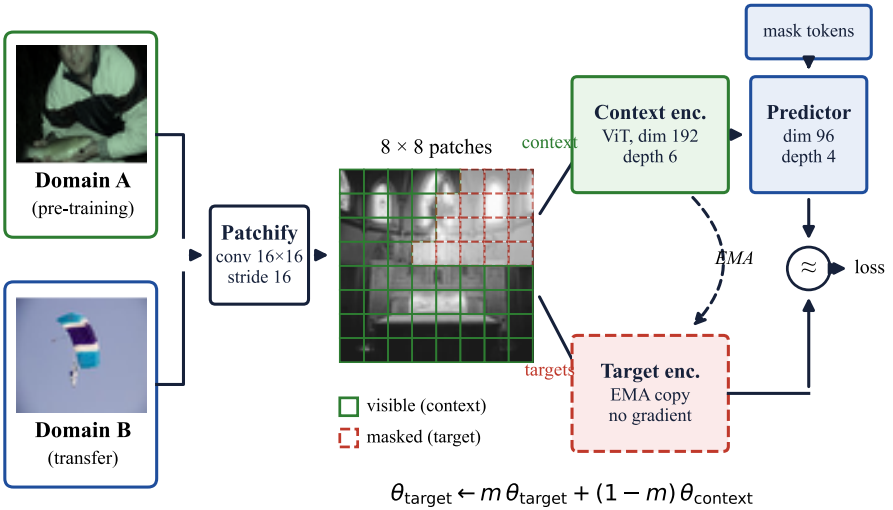


FIG 17 *Method diagram for masked-feature prediction. Two input domains are split into 8 × 8 patches; visible patches go to the context encoder and masked ones to a target encoder updated by exponential moving average (dashed, no gradient). The predictor's output is matched to the targets. Real photos (Imagenette); numbers are illustrative.*

USE IT WHEN

you introduce a method with more than two moving parts.

HOW TO BUILD IT

- Left to right in the order data flows; one colour per role (context, target, predictor).
- Dashed outlines and arrows for things that receive no gradient or are updated differently.
- Real input images, not icons, so readers see the data.
- Put the one key equation under the diagram.

AVOID

- Software class names or file names as labels.
- Arrows that cross when the order could be changed.
- Text smaller than the caption font.

6. The annotated screenshot: the system in use

For systems papers, a screenshot proves the thing exists. Annotate only what the text refers to, in the paper's font, so the labels read as part of the figure rather than part of the user interface.

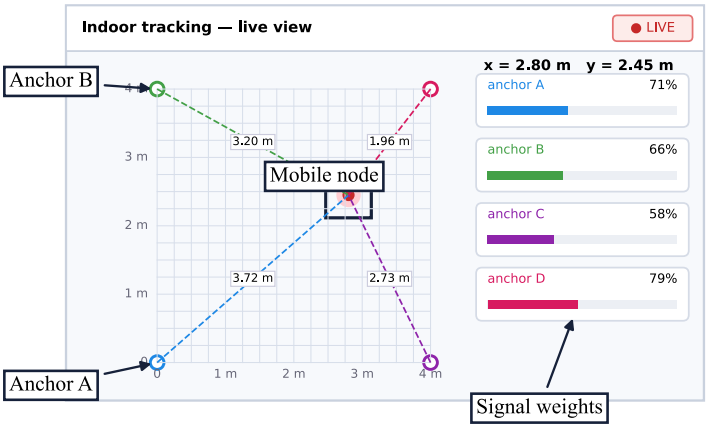


FIG 18 *Annotated live view of an indoor tracking system. Callouts in the paper's typeface mark the anchors, the tracked node and the signal-weight panel; everything else is left as the interface shows it. Illustrative data.*

USE IT WHEN

the interface, hardware or output is part of the contribution or the evaluation.

HOW TO BUILD IT

- Capture at high resolution, then crop to what matters.
- Boxed callouts in the paper's serif font, with thin arrows.
- Blank out personal data, IP addresses and tokens.
- Name the callouts exactly as the text does.

AVOID

- Callouts that cover the data they label.
- Browser chrome and desktop clutter.
- Labelling every widget.

7. The system diagram: hardware and data path

When a study involves devices, networks or several programs, one diagram should show what runs where and how data moves between them, with a legend for the kinds of link.

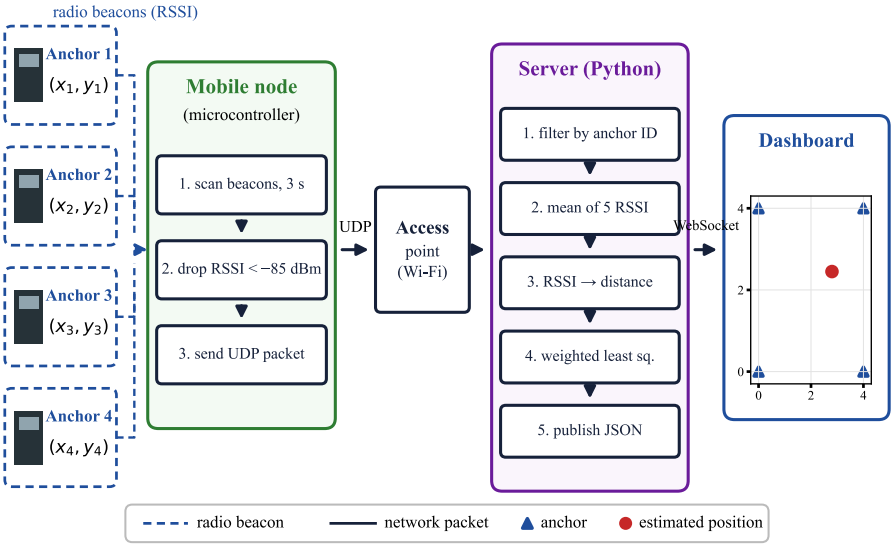


FIG 19 *System diagram: anchors broadcast radio beacons; the mobile node filters them and sends packets through an access point; the server turns signal strength into a position estimate and streams it to a dashboard. Dashed and solid lines distinguish radio and network links. Illustrative data.*

USE IT WHEN

the method depends on hardware, networking or several cooperating programs.

HOW TO BUILD IT

- Group by device; inside each device, number the processing steps.
- A legend for each kind of link (radio, network, file).
- Give the key parameters on the diagram (thresholds, rates, ports) so the text can stay short.
- Match colours with the screenshot and method figure.

AVOID

- Brand logos and clip art.
- Showing every library used.
- Arrows without a meaning in the legend.

BEFORE YOU SUBMIT, CHECK THE SET

Every figure is cited in the text before it appears. One typeface and one colour scheme across all figures. Panel labels instead of titles. Captions that say what is shown, under which conditions, and what to notice. Selection rules stated for every grid of examples.

CHAPTER 9

Discussion, Limits & Conclusion

Interpretation, honest limits, and a conclusion that answers the questions you asked.

Discussion: result → interpretation → implication

Do not repeat the results. Explain **what they meant, why they occurred, why one approach outperformed another, how they relate to what was known, and what limitations affect interpretation.**

Close every research question

A reader should be able to trace each question through the paper in one line. This is the most persuasive structure you can give a reviewer.

RQ	Evidence	Finding	Answer
RQ1	Table 2, Fig. 4	Error fell under shifted conditions	Yes, within the tested range
RQ2	Fig. 5, ablation	Removal of component B erased the gain	B is responsible for the gain
RQ3	Table 4	Cost scaled linearly	Feasible on modest hardware

Illustrative only. In your paper, every row points to an actual table or figure.

Limitations are a strength when stated plainly

- Simulation-only, or lab-only validation
- Few scenarios or narrow conditions
- Dataset or sample limitations
- Model assumptions
- Hardware or compute constraints
- Sensor / measurement limits
- No real-world or external validation
- Limited generalisation

State them objectively. Do not hide meaningful limits, and do not artificially undermine a strong study.

The conclusion

Synthesise the answers to the research questions and the contribution without exaggeration. Do not paste the abstract again, introduce new results, or promise what the data cannot support. End on the implication and the clearest next step.

> **READ IT ONCE AS A STORY**
>
> Can you read the paper as one continuous story, a real problem; a gap in prior work; specific questions; a justified approach; a controlled experiment; measurable evidence; clear visuals; answers; an explanation; stated limits; a measured conclusion, with no section that feels stapled on afterward?

CHAPTER 10

The Human–AI Workflow

Treat the AI as a quick junior colleague who never tires and never takes responsibility. You supply the facts and you sign your name.



FIG 20 *The five-step loop. Verification is a step of its own, never an afterthought.*

What goes into a good brief

- **Role and goal:** which section, for which venue and audience.
- **Ground truth:** your real results, tables, parameters and notes. The AI should not invent any.
- **Rules:** the style rules from Chapters 2–8, pasted or summarised.
- **Boundaries:** "do not add citations", "do not introduce new numbers", "flag anything uncertain".
- **Output form:** prose, an outline, or a list of problems (not always a rewrite).

PROMPT CARD **SECTION BRIEF**

```
You are an academic researcher, scientific writer and peer reviewer in
[FIELD].
Task: draft the [SECTION] of a paper for [VENUE].
Use ONLY the facts below. Do not invent numbers, results or references.
If something is missing, list it as a question instead of guessing.

FACTS: [paste results, parameters, notes]
RULES: third person; past tense for completed work; no marketing language;
no code or function names; explain procedures as algorithms; cite only
where I mark [CITE]; one term per concept.
OUTPUT: the section text, then a list of claims you were unsure about.
```

Where AI goes wrong, and the guard

Failure mode	Guard
Invented or mangled references	Treat every citation as unverified until you open the source. Never accept a reference you cannot find.
Fabricated or "rounded" numbers	Provide the data; check every figure against the table. Ask for no new values.
Overclaiming ("proves", "revolutionary")	Apply the claim ladder; require hedged verbs matched to evidence.
Report-style drift (code, files, "we ran")	Ask for algorithmic explanation; ban function and file names.
Repetition, padding, term drift	Editor pass: "remove redundancy, keep every fact once"; supply a glossary with one term per concept.
Smooth text hiding weak logic	Run a reviewer pass that attacks the argument, not the grammar.

INTEGRITY

You are accountable for everything in the manuscript. Follow your journal's and institution's policy on disclosing AI assistance; never present machine-generated references, data or figures as real evidence.

CHAPTER 11

The Reviewer's Checklist

Read the paper once more as a skeptical reviewer would. Twenty questions follow; each "no" means more revision.

- [ ] Is the research problem clear?
- [ ] Is the research gap genuine?
- [ ] Are the RQs clearly stated?
- [ ] Does the methodology answer the RQs?
- [ ] Are methodological decisions justified?
- [ ] Is the experimental design sufficient?
- [ ] Are the baselines appropriate?
- [ ] Are the metrics appropriate?
- [ ] Do the results support the claims?
- [ ] Are the figures scientifically useful?
- [ ] Is every table necessary?
- [ ] Are limitations acknowledged?
- [ ] Is the work reproducible?
- [ ] Is the narrative coherent?
- [ ] Is there unnecessary repetition?
- [ ] Are any claims unsupported?
- [ ] Does the conclusion answer the RQs?
- [ ] Is it a paper, not a technical report?
- [ ] Could a reviewer misread a key decision?
- [ ] Does every major claim have evidence?

Final output checks

- [ ] Third person, no personal pronouns
- [ ] Past tense where work is described
- [ ] Abstract self-contained, no citations
- [ ] RQs labelled and all answered
- [ ] Introduction ends with road map
- [ ] Every section opens, closes, transitions
- [ ] No code, files or functions in prose
- [ ] Hyperparameters in one place
- [ ] Figures introduced, interpreted, captioned
- [ ] Numbers stated once
- [ ] Terms and units consistent
- [ ] Claims match the evidence ladder

Three review passes worth running

Logic pass

"Act as a skeptical reviewer. List the three weakest links in the argument chain and the evidence that would fix each."

Evidence pass

"Find every claim that exceeds its evidence. Quote it, and suggest a proportionate verb."

Style pass

"Find passive/active imbalance, report-style phrasing, number repetition and undefined acronyms. Do not rewrite."

Master prompt

COPY & PASTE **MANUSCRIPT RULES**

```
Act simultaneously as an academic researcher, scientific writer, peer
reviewer, domain expert and critical editor. Write/revise this manuscript
as a research paper, never as a technical or project report.

1. Third person only; past tense for completed work, present for facts.
2. Formal, precise tone; no promotional language; claims proportional
   to evidence.
3. Story: Problem > Gap > RQs > Approach > Experiment > Evidence >
   Interpretation > Answer > Implication.
4. Label RQ1..RQn; answer each explicitly; end the introduction with
   the paper structure.
5. Abstract: self-contained, no citations.
6. Literature review: synthesise by theme toward the gap.
7. Describe science, not software: no function/file names; explain
   logic as algorithms or equations.
8. Methods reproducible; primary method first, others as baselines.
9. Report results before interpreting; state each number once; give
   uncertainty; never claim more than the evidence shows.
10. Every figure introduced, captioned and interpreted; one typeface.
11. Discussion explains why; limitations stated; conclusion resolves
    the RQs.
Finally, review the manuscript as a skeptical referee and list every
weakness before declaring it complete. Do not invent data or citations.
```

CHAPTER A

Appendix: The Full Rulebook

The complete reference text this guide was built from. The UAV and quadrotor rules are kept as written; read them as an example of how to make the general rules concrete for one field.

Comprehensive Academic Research Paper Writing Rules for AI

1. Core Objective

The manuscript must be written as a **fully academic research paper**, not as a technical report, project report, software documentation, implementation report, or thesis chapter.

The paper must read like a coherent scientific investigation. The reader should understand:

- What problem was studied.
- Why the problem mattered.
- What gap existed in existing research.
- What research questions were investigated.
- How the problem was addressed.
- Why the selected methodology was appropriate.
- What was found.
- What the findings meant.
- How the findings related to existing research.
- What limitations existed.
- What conclusions could reasonably be drawn.

The manuscript must prioritize **scientific reasoning, evidence, interpretation, and narrative coherence** over implementation details.

The AI must behave simultaneously as:

1. An academic researcher.
2. A scientific writer.
3. A peer reviewer.
4. A domain expert.
5. A critical editor.

The AI must not merely rewrite sentences. It must identify weaknesses in logic, structure, scientific explanation, methodology presentation, evidence, visualization, and interpretation.

2. Academic Tone

The entire manuscript must maintain a formal academic tone.

The writing must be:

- Precise.
- Objective.
- Evidence-driven.
- Concise.
- Grammatically correct.
- Scientifically defensible.
- Easy to follow.
- Free from conversational language.

Avoid:

- Marketing language.
- Promotional claims.
- Unnecessary adjectives.
- Informal expressions.
- Personal opinions.
- Unsupported claims.
- Dramatic statements.
- Project-report language.
- Excessive implementation terminology.

Words such as "very", "excellent", "amazing", "powerful", "highly effective", "revolutionary", and "significant" must not be used unless they are scientifically justified by evidence.

Claims must be proportional to the evidence.

---

3. Person and Perspective

The paper must be written entirely in the **third person**.

Do not use:

- I
- We
- Us
- Our
- My
- Me

Instead, use constructions such as:

- "The study investigated..."
- "The proposed framework employed..."

- "The experiment demonstrated..."
- "The results indicated..."
- "The analysis revealed..."

The authors must remain invisible in the prose unless the target journal explicitly requires another convention.

---

4. Verb Tense

The manuscript should primarily use **past tense** because it describes completed research.

Use past tense when describing:

- What the study did.
- What methodology was employed.
- What experiments were conducted.
- What datasets were used.
- What simulations were performed.
- What results were obtained.
- What observations were made.

Examples:

"The proposed framework employed a quadrotor model."

"The simulation was conducted under two environmental scenarios."

"The CNN achieved the highest classification accuracy."

Present tense may be used selectively for:

- Established scientific facts.
- General principles.
- Definitions.
- Mathematical truths.
- Statements about the structure of the paper.

For example:

"Quadrotor dynamics are affected by aerodynamic forces."

"Section 4 presents the experimental results."

Do not force every sentence into past tense when doing so would make established scientific statements grammatically unnatural.

---

5. Active and Passive Voice

The manuscript must contain an appropriate mixture of active and passive constructions.

Do not write the entire paper in passive voice.

Do not write the entire paper in active voice either.

Use active voice when it improves clarity:

"The analysis identified three dominant factors."

Use passive voice when the process or object is more important than the actor:

"The simulation environment was configured using the selected parameters."

The primary objective is **clarity**, not mechanical adherence to active/passive ratios.

---

6. Paper Must Not Read Like a Technical Report

Avoid project-report terminology such as:

- "The code was written..."
- "The program was executed..."
- "The function was called..."
- "The script was run..."
- "The software was developed..."
- "The user clicked..."
- "The system printed..."
- "The following function..."
- "The following code..."
- "The implementation consisted of..."

Instead, describe the scientific operation.

For example:

Instead of:

"The Python function calculated the velocity."

Write:

"The velocity was calculated from the measured positional states."

Instead of:

"The code used a CNN function to classify the images."

Write:

"A convolutional neural network was employed to extract hierarchical spatial features and perform classification."

The paper must explain **what was scientifically done and why**, rather than describing how individual programming components were implemented.

---

7. No Direct Code-Function References

Programming functions, classes, variables, filenames, modules, source-code structures, and implementation-specific function calls must not be directly discussed unless they are scientifically essential.

Do not write:

"Function calculate_velocity() was used."

"Algorithm.py contained the preprocessing function."

"Model.predict() generated the output."

Instead, describe the underlying operation:

"The velocity was estimated from the positional measurements."

"The preprocessed observations were subsequently passed to the trained model."

"Predictions were generated using the trained classifier."

The preferred approach is **algorithmic explanation**.

---

8. Algorithmic Explanation

When explaining computational logic, algorithms, workflows, or procedures, prefer:

- Mathematical formulations.
- Pseudocode.
- Algorithm blocks.
- Process descriptions.
- Equations.
- Flow diagrams.

Do not replace scientific explanation with programming implementation details.

If a process requires multiple logical steps, present it as an algorithm.

The algorithm must explain:

1. Input.
2. Initialization.
3. Processing.
4. Decision conditions.
5. Iteration where applicable.
6. Output.
7. Termination condition where applicable.

The algorithm must be understandable without access to the source code.

---

9. Abstract

The abstract must be completely self-contained.

The abstract must not contain references or citations.

The abstract should communicate:

1. Background/problem.
2. Research gap or motivation.
3. Objective.
4. Methodology.
5. Main results.
6. Main implication/conclusion.

The abstract must not:

- Introduce unexplained acronyms.
- Contain citations.
- Contain excessive technical implementation details.
- Repeat the entire methodology.
- Make claims unsupported by the paper.
- Contain unnecessary numerical results.

The abstract must provide enough information for a reader to understand what was investigated and what was discovered.

---

10. Introduction

The introduction must establish the research story progressively.

A strong introduction should generally move through:

Context → Problem → Importance → Existing limitation → Research gap → Research objective → Research questions → Contributions → Paper structure

The introduction must answer:

- What is the broader problem?
- Why does it matter?
- What has already been done?
- What remains unresolved?
- Why is the unresolved problem important?
- What does this study investigate?
- How does the study address the gap?

The introduction must not become a detailed literature review.

Technical implementation details should not dominate the introduction.

---

11. Research Questions

Research questions must be explicitly identified using the format:

RQ1: ...

RQ2: ...

RQ3: ...

Each research question must be:

- Clearly stated.
- Specific.
- Researchable.
- Answerable using the presented methodology.
- Directly connected to the results.

Every RQ must eventually be answered in the paper.

No research question should be introduced merely for appearance.

The results and discussion must make it possible for a reviewer to identify where and how each RQ was answered.

---

12. Introduction Ending

The introduction must end with a concise description of the paper structure.

For example:

"The remainder of the paper is organized as follows. Section 2 reviews the relevant literature and identifies the research gap. Section 3 presents the proposed methodology. Section 4 describes the experimental environment and evaluation protocol. Section 5 presents and analyzes the results. Section 6 discusses the findings and limitations. Section 7 concludes the study."

The exact structure must reflect the actual manuscript.

Do not describe sections that do not exist.

---

13. Literature Review

The literature review must be analytical rather than descriptive.

Do not simply summarize papers one after another.

Avoid the pattern:

"Author A did X. Author B did Y. Author C did Z."

Instead, synthesize literature around:

- Research themes.
- Methodological approaches.

- Strengths.
- Limitations.
- Contradictions.
- Trends.
- Unresolved problems.
- Methodological gaps.

The literature review must progressively lead toward the research gap.

Each paragraph should contribute to the argument.

The literature review should answer:

- What is already known?
- What approaches have been used?
- What are their limitations?
- What remains unresolved?
- Where does the present study fit?

---

14. Citation Boundary

References/citations must not appear in:

- Abstract.
- Methodology.
- Experimental setup.
- Results.
- Discussion.
- Conclusion.

References should primarily support claims in:

- Introduction.
- Literature review.

However, this rule applies only where the target publication venue permits such citation practice. If a scientific claim in a later section genuinely requires external attribution, the manuscript must follow the target journal/conference's citation requirements rather than artificially removing necessary attribution.

No citation should be inserted merely to make a paragraph appear academic.

---

15. Section Transitions

Every major section must have:

1. A meaningful opening.
2. A logical development.

3. A meaningful conclusion.
4. A transition to the next section.

Sections must not feel like independent documents placed together.

For example, the methodology should naturally emerge from the research gap.

The experimental setup should naturally emerge from the methodology.

The results should naturally emerge from the experimental design.

The discussion should naturally emerge from the results.

The conclusion should naturally emerge from the discussion.

The manuscript must feel like **one continuous scientific story**.

---

16. Narrative Coherence

The paper must be easy to read from beginning to end.

The logical chain should resemble:

Problem → Gap → Questions → Approach → Experiment → Evidence → Interpretation → Answer → Implication

Every major section should strengthen this chain.

Avoid sudden jumps between topics.

Avoid introducing terminology that has not been established.

Avoid presenting results before explaining what they measure.

Avoid introducing methodology that is not motivated by the research question.

Avoid presenting figures or tables without explaining why they matter.

---

17. Every WH Question Must Be Answered

The manuscript must collectively answer the major WH questions:

What was studied?

Why was it studied?

Where was it studied?

When was it studied, where relevant?

Who/what constituted the experimental subject, system, dataset, or model?

How was the problem investigated?

How much data, computational resources, or experimental material was used?

Which methods, parameters, scenarios, or models were selected?

Why was each important methodological decision made?

What happened during the experiments?

What did the results mean?

Why did the observed results occur?

What are the implications?

If an important question naturally arises while reading the paper and the manuscript does not answer it, the manuscript is incomplete.

---

18. Methodology

The methodology must explain the scientific procedure sufficiently for reproducibility without becoming a software manual.

It should clearly describe:

- Research design.
- System architecture.
- Experimental environment.
- Hardware.
- Software environment where scientifically relevant.
- Simulation environment.
- Dataset or experimental input.
- Variables.
- Parameters.
- Model configuration.
- Hyperparameters.
- Evaluation metrics.
- Experimental scenarios.
- Baselines/comparison methods.
- Experimental protocol.

The methodology must answer:

What was done?

Why was it done?

How was it done?

Under what conditions was it done?

---

19. Multiple Methodologies

If several candidate methodologies were investigated, the paper must not give equal narrative emphasis to all of them.

The strongest methodology must be presented as the primary approach.

Other methods may be presented as:

- Baselines.
- Comparative methods.
- Ablation alternatives.
- Benchmark approaches.

The manuscript must clearly establish why the primary methodology was selected.

The paper should not resemble an implementation diary in which every attempted method receives equal space.

---

20. Comparison Studies

Multiple methodologies may be compared when scientifically useful.

However, comparisons must focus on:

- Performance.
- Scientific relevance.
- Robustness.
- Computational cost.
- Generalization.
- Suitability for the research problem.
- Strengths and weaknesses.

Do not turn the comparison into a technical-report inventory.

The comparison must answer:

Which approach performed better, under what conditions, and why?

---

21. Hardware and Simulation Papers

For hardware-, robotics-, control-, or simulation-oriented research, the paper must provide sufficient information to understand and reproduce the experimental environment.

At minimum, the relevant manuscript should include:

- Overall system architecture.
- Hardware components.
- Sensor configuration.
- Actuator configuration.
- Computational platform.
- Communication interfaces where relevant.
- Simulation environment.
- Simulation parameters.

- Quadrotor model.
- Quadrotor specifications.
- Environmental/scenario configuration.
- Controller or algorithm configuration.
- Hyperparameters.
- Evaluation criteria.

Only scientifically relevant specifications should be reported.

Do not dump complete hardware datasheets into the paper.

---

22. Quadrotor Model

The **final Quadrotor model must be used as the primary visual model throughout the paper**.

The model must remain visually consistent across:

- System architecture.
- Simulation environment.
- Scenario illustrations.
- Experimental figures.
- Model screenshots.
- Result visualizations.

Do not use different quadrotor representations in different figures unless the difference has a scientific purpose.

The quadrotor section must explain:

- Model type.
- Configuration.
- Number and arrangement of rotors.
- Coordinate frame.
- Relevant dimensions.
- Mass.
- Inertia where relevant.
- Propulsion configuration.
- Sensors.
- Actuators.
- Flight-control components.
- Simulation-specific properties.
- Any assumptions.

Only parameters relevant to the study should be included.

---

23. Two Experimental Scenes

The two experimental scenes must be explicitly described.

For each scene, explain:

- Purpose.
- Environment.
- Geometry.
- Obstacles.
- Initial conditions.
- Quadrotor starting position.
- Target position.
- Relevant environmental features.
- Disturbances, if any.
- Navigation/control challenge.
- Why the scene was included.
- What scientific question the scene helped evaluate.

The scenes must not merely be presented as screenshots.

The text must explain what each scene tested.

The differences between the scenes must be scientifically meaningful.

---

24. System Architecture

A system architecture figure should show the major scientific components and their relationships.

A typical architecture may contain:

Input/Environment → Sensors → State Estimation → Control/Decision Layer → Actuation → Quadrotor Dynamics → Environment Feedback

The exact architecture must reflect the actual system.

The architecture must not become a software class diagram.

Avoid showing:

- Source-code functions.
- File names.
- Programming variables.
- Unnecessary implementation modules.

The architecture must communicate the system's **scientific information flow**.

---

25. Simulation Environment

The simulation environment must be described clearly enough that another researcher could understand the experimental conditions.

Include relevant:

- Simulator.
- Physics engine.
- World/environment.
- Model configuration.
- Sensor configuration.
- Actuator configuration.
- Time step.
- Simulation duration.
- Initial conditions.
- Boundary conditions.
- Environmental assumptions.
- Disturbance conditions.
- Control frequency.
- Evaluation conditions.

Do not list software simply because it was installed.

Only include tools and configurations relevant to reproducibility.

---

26. Hyperparameters

All scientifically relevant hyperparameters must be reported.

Depending on the system, this may include:

- Learning rate.
- Batch size.
- Number of epochs.
- Optimizer.
- Network depth.
- Number of filters.
- Kernel size.
- Activation functions.
- Dropout.
- Regularization.
- Controller gains.
- Sampling time.
- Control frequency.
- Population size.

- Iteration count.
- Search parameters.
- Thresholds.
- Simulation step size.

Hyperparameters must be presented in a compact and readable manner.

Do not repeatedly mention the same values throughout the paper.

---

27. CNN Architecture

If a CNN is used, the architecture must be explicitly documented.

The paper must state, where applicable:

- Input dimensions.
- Number of convolutional layers.
- Number of filters per layer.
- Kernel sizes.
- Strides.
- Padding.
- Activation functions.
- Pooling layers.
- Flattening operation.
- Fully connected layers.
- Output layer.
- Dropout or regularization.
- Number of trainable parameters where relevant.

A CNN architecture figure should be provided when the architecture is scientifically important.

The figure should allow a reader to understand the transformation:

Input → Feature Extraction → Feature Compression → Classification/Regression Output

Do not simply state that "a CNN was used."

---

28. Mathematical Formulation

Where mathematical models are central to the study, the underlying equations must be provided.

Equations should be introduced before they are used.

Every important variable must be defined.

For a quadrotor/control study, relevant formulations may include:

- Position dynamics.
- Translational dynamics.
- Rotational dynamics.

- Attitude representation.
- Control inputs.
- Forces and moments.
- State-space representation.
- Controller formulation.
- Error formulation.
- Objective/cost function.

Do not introduce equations merely for decoration.

Every equation must serve a scientific purpose.

---

29. Results

The results section must report findings objectively before interpreting them extensively.

Results should answer:

- What was observed?
- How large was the effect?
- Under which condition?
- How did methods differ?
- How consistent were the findings?
- What evidence supports the finding?

The results must not simply reproduce raw logs.

---

30. Numerical Results

Numbers must not be unnecessarily repeated throughout the manuscript.

If a value has already been clearly reported in a table or figure, the surrounding text should focus on its meaning rather than repeating every number.

For example, instead of repeatedly stating:

"Method A achieved 95.2%, while Method B achieved 92.1%..."

the paper should state the exact values once and then discuss:

"Method A consistently outperformed the baseline, indicating improved predictive capability under the evaluated conditions."

Use numbers when they strengthen the argument.

Do not use numbers merely to make the prose appear quantitative.

---

31. Statistical and Experimental Evidence

Where applicable, report:

- Mean.
- Standard deviation.
- Confidence intervals.
- Error margins.
- Statistical significance.
- Number of trials.
- Variability.
- Repeated experiments.

Do not make claims of improvement without sufficient evidence.

If a result is based on a single simulation or experiment, do not describe it as statistically robust.

---

32. Visualization Philosophy

Visualizations must communicate information **immediately**.

A reader should understand the main message of a figure within a few seconds.

Every visualization must have a clear purpose.

Avoid:

- Decorative figures.
- Overcrowded plots.
- Tiny labels.
- Excessive colors.
- Unnecessary 3D effects.
- Redundant graphs.
- Figures containing too much information.
- Ambiguous legends.

A figure should answer a specific question.

---

33. Visualization Quality

All figures must be publication-quality.

Use:

- High resolution.
- Consistent typography.
- Consistent terminology.
- Clear axis labels.
- Meaningful legends.
- Appropriate scale.
- Adequate whitespace.

- Consistent dimensions.
- Consistent visual language.

All visualization text must use **Times New Roman** unless the target publication explicitly requires another font.

Figure labels, axis labels, annotations, legends, and titles should use Times New Roman.

---

34. Visualization Selection

Use the simplest visualization that communicates the result correctly.

Examples:

Trajectory comparison → 2D/3D trajectory plot.

Position error → line plot.

Control response → time-series plot.

Method comparison → bar/box/violin plot depending on the data.

Distribution → histogram/density plot.

System workflow → architecture diagram.

Quadrotor environment → annotated simulation screenshot.

CNN structure → architecture diagram.

Parameter sensitivity → heatmap or line plot.

Do not use a complicated visualization when a simpler figure communicates the same information better.

---

35. Figures Must Be Explained

Every figure must be introduced in the text before or at the point where it becomes relevant.

Do not write:

"Figure 7 shows the result."

Instead explain:

"Figure 7 compares the positional trajectories obtained under the two environmental conditions, allowing the stability and path-following behavior of the proposed controller to be evaluated."

The figure caption must describe what the reader is looking at.

The surrounding paragraph must explain **why the figure matters**.

---

36. Figure Captions

Captions must be self-contained and informative.

A good caption should allow a reader to understand the figure without reading the entire paragraph.

Avoid:

"Simulation result."

Prefer:

"Comparison of quadrotor trajectories under the two simulated environmental conditions, showing the reference path and the trajectory generated by the proposed controller."

---

37. Quadrotor Visualization Prompts

Whenever an architectural diagram, simulation environment, quadrotor model, or scenario figure must be generated using an external image-generation system such as Nano Banana, provide a **detailed generation prompt**.

The prompt must specify:

- Exact subject.
- Camera angle.
- Environment.
- Objects.
- Object positions.
- Relevant labels.
- Scientific purpose.
- Required visual style.
- Background.
- Resolution.
- Perspective.
- Lighting.
- Level of detail.
- Publication-quality requirements.
- Label placement.
- Font.

Every generated scientific figure prompt must explicitly specify:

"Use Times New Roman for all labels, annotations, legends, and textual elements."

The generated figure must not contain:

- Decorative text.
- Fake equations.
- Unnecessary labels.
- Unreadable annotations.
- Random UI elements.
- Watermarks.

- Artistic distortions that compromise scientific interpretation.

38. Quadrotor Model Screenshot Generation

For the final quadrotor model, generate high-quality screenshots suitable for direct use as paper figures.

The screenshot prompts must specify:

- The exact final quadrotor model.
- Full model visibility.
- High-resolution rendering.
- Clean background.
- Appropriate camera orientation.
- Orthographic or controlled perspective view where appropriate.
- Clearly visible rotor arrangement.
- Clearly visible body.
- Clearly visible sensors.
- Clearly visible relevant components.
- No unnecessary objects.
- No distracting interface elements.

If labeling is required, labels must identify only scientifically relevant components.

All labels must use **Times New Roman**.

The final screenshots must visually correspond to the model actually used in the experiments.

39. Tables

Tables must be used when they communicate structured information more efficiently than prose.

Appropriate table content includes:

- Hardware specifications.
- Quadrotor parameters.
- Hyperparameters.
- Experimental scenarios.
- Evaluation metrics.
- Comparison results.

Do not create tables merely because information can technically be placed into rows and columns.

Tables must have:

- Clear titles.
- Meaningful column headings.
- Consistent units.

- Appropriate decimal precision.
- Explanatory notes where necessary.

---

40. Units and Notation

Scientific units must be consistent throughout the paper.

Do not alternate between equivalent representations unnecessarily.

For example, if velocity is reported in m/s, do not switch between m/s and km/h unless the comparison requires it.

Symbols must remain consistent.

The same variable must not represent different quantities in different sections.

---

41. Abbreviations

Every abbreviation must be defined at first use.

For example:

"Computational Fluid Dynamics (CFD)"

After definition, use the abbreviation consistently.

Do not repeatedly redefine abbreviations.

Avoid unnecessary abbreviations.

---

42. Discussion

The discussion must not merely repeat the results.

It must explain:

- What the results meant.
- Why the results occurred.
- How the findings related to the research questions.
- Why one methodology performed better.
- What the findings implied.
- Where the findings agreed or disagreed with existing knowledge.
- What limitations affected interpretation.

The discussion must move from **result → interpretation → scientific implication**.

---

43. Results Versus Discussion

Maintain a clear distinction.

Results: What happened?

Discussion: Why did it happen, and what does it mean?

Do not place extensive interpretation into the results section.

Do not introduce completely new experimental results in the discussion.

---

44. Research Question Resolution

The paper must explicitly resolve every RQ.

A reader should be able to identify:

RQ1 → Evidence → Finding → Answer

RQ2 → Evidence → Finding → Answer

RQ3 → Evidence → Finding → Answer

The conclusion should synthesize these answers rather than merely repeating the abstract.

---

45. Limitations

The paper must acknowledge genuine limitations.

Potential limitations may include:

- Simulation-only validation.
- Limited environmental complexity.
- Limited number of scenarios.
- Hardware constraints.
- Sensor limitations.
- Computational constraints.
- Dataset limitations.
- Model assumptions.
- Absence of real-world validation.
- Limited generalization.

Limitations must be stated objectively.

Do not artificially weaken a strong study, but do not hide meaningful limitations.

---

46. Claims and Evidence

Every major scientific claim must be supported by one of:

- Experimental evidence.
- Simulation evidence.
- Mathematical analysis.
- Statistical analysis.
- Established literature.

Do not make claims that exceed the evidence.

For example, simulation results should not automatically be described as proof of real-world performance.

Use precise language such as:

"The simulation results indicated..."

rather than:

"The system was proven to work in all real-world conditions."

---

47. Reproducibility

The methodology must provide sufficient information for another researcher to reproduce the experiment.

Reproducibility should include, where relevant:

- Dataset/source.
- Experimental conditions.
- Model configuration.
- Hardware.
- Simulation environment.
- Hyperparameters.
- Initial conditions.
- Evaluation metrics.
- Number of trials.
- Randomization settings.
- Comparison methods.

Reproducibility must be achieved without turning the manuscript into source-code documentation.

---

48. Avoid Redundancy

The same information should not be explained repeatedly.

If a parameter is fully defined in the methodology, do not redefine it in the results.

If a figure already communicates the exact values, do not repeat every value in the paragraph.

If a table contains all specifications, discuss only the specifications relevant to the argument.

The manuscript must be concise without becoming incomplete.

---

49. Terminology Consistency

Use the same terminology throughout the paper.

Do not alternate unnecessarily between:

- Quadrotor / drone / UAV / aircraft.
- Controller / control algorithm / control system.
- Environment / scene / scenario.

Different terms may be used only when they genuinely represent different concepts.

---

50. Section-Level Quality

Each major section must satisfy four conditions:

Opening: Why is this section necessary?

Body: What evidence or information does it provide?

Closing: What has been established?

Transition: Why does the next section logically follow?

No section should end abruptly.

---

51. Paragraph-Level Quality

Every paragraph should have a clear purpose.

A strong paragraph should generally contain:

Topic → Evidence/Explanation → Interpretation → Connection

Avoid paragraphs containing unrelated facts.

Avoid one-sentence paragraphs unless they serve a deliberate structural purpose.

Avoid extremely long paragraphs containing several independent arguments.

---

52. Sentence-Level Quality

Sentences must be:

- Grammatically correct.
- Direct.
- Precise.
- Scientifically meaningful.

Avoid unnecessarily complicated sentence structures.

Do not use sophisticated vocabulary merely to sound academic.

Academic writing should be **clear, not artificially complex**.

---

53. Reviewer-Level Critical Check

Before finalizing the manuscript, the AI must internally evaluate the paper as a peer reviewer.

It must check:

1. Is the research problem clear?
2. Is the research gap genuine?
3. Are the RQs clearly stated?
4. Does the methodology answer the RQs?
5. Are methodological decisions justified?
6. Is the experimental design sufficient?
7. Are the baselines appropriate?
8. Are the reported metrics appropriate?
9. Are the results sufficient to support the claims?
10. Are the figures scientifically useful?
11. Are the tables necessary?
12. Are the limitations acknowledged?
13. Is the paper reproducible?
14. Is the narrative coherent?
15. Is there unnecessary repetition?
16. Are unsupported claims present?
17. Does the conclusion answer the RQs?
18. Does the manuscript resemble a research paper rather than a technical report?
19. Could a reviewer misunderstand any important methodological decision?
20. Does every major claim have adequate evidence?

If any answer is negative, the manuscript must be revised before being considered complete.

---

54. Final Paper Story

The final manuscript must feel like one continuous story:

A real problem existed.

Existing research addressed parts of the problem but left a meaningful gap.

That gap motivated specific research questions.

The selected methodology was designed to answer those questions.

The experimental environment was carefully established.

The quadrotor model and two scenes provided controlled evaluation conditions.

The experiments generated measurable evidence.

The visualizations made the evidence immediately understandable.

The results answered the research questions.

The discussion explained why the results occurred and what they meant.

The limitations established the boundaries of the conclusions.

The conclusion summarized the scientific contribution without exaggeration.

The reader should never feel that unrelated sections were assembled afterward.

---

55. Final Output Requirements

Before producing the final manuscript, the AI must ensure that:

- The manuscript is fully academic.
- The manuscript is written in third person.
- Personal pronouns are absent.
- Past tense is used appropriately.
- Active and passive voice are balanced naturally.
- The abstract contains no references.
- Citations are concentrated in the introduction and literature review unless publication requirements require otherwise.
- RQs are explicitly labeled RQ1, RQ2, etc.
- Every RQ is answered.
- The introduction ends with the paper structure.
- Every section has a beginning, development, ending, and transition.
- The manuscript reads as one coherent story.
- The manuscript does not resemble a technical report.
- Code functions are not discussed directly.
- Algorithms are preferred for explaining computational logic.
- The strongest methodology receives primary attention.
- Alternative methodologies are used mainly for comparison where appropriate.
- The CNN architecture is explicitly described when applicable.
- Hardware and simulation details are sufficiently documented.
- The system architecture is included where appropriate.
- The simulation environment is clearly described.
- The final quadrotor model is used consistently.
- Both experimental scenes are clearly described.
- Quadrotor specifications are reported.
- Relevant hyperparameters are reported.
- Visualizations are publication-quality and immediately interpretable.
- Figures are not decorative.
- Every figure is introduced and interpreted.
- Figure captions are informative.
- Visualization prompts are provided when external image generation is required.
- Visualization prompts explicitly specify Times New Roman.
- High-quality labeled screenshots of the final quadrotor model are generated where required.
- Numerical results are not unnecessarily repeated.

- Claims do not exceed the evidence.
- Limitations are acknowledged.
- Terminology is consistent.
- The paper is reproducible.
- The conclusion directly resolves the research questions.
- The entire manuscript has been checked from the perspective of a skeptical academic reviewer.

Golden Rule

The manuscript must never answer the question:

"How was the software implemented?"

unless implementation itself is the research contribution.

Instead, it must answer:

"What scientific problem was investigated, why was the selected approach appropriate, how was it evaluated, what evidence was obtained, and what did that evidence demonstrate?"

The final paper must prioritize **scientific contribution over implementation detail, interpretation over description, evidence over claims, and coherence over information density.**

THE AUTHOR

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